Hall Ticket Number:

MR-21

Code: B314CS5S3

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, December 2024/January 2025

Discrete Mathematics

(Common to CSE, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

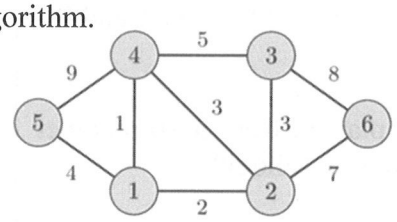
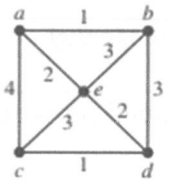
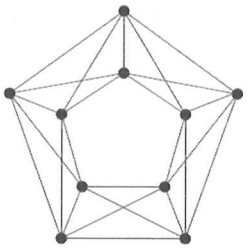
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.	2	1	1	1,2
1. b)	Let P (x) be the statement "x can speak Russian" and let Q(x) be the statement "x knows the computer language C++." Express each of these sentences in terms of P (x), Q(x), quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school. (i) There is a student at your school who can speak Russian and who knows C++ (ii) Every student at your school either can speak Russian or knows C++.	2	2	1	1,2
1. c)	Draw a Venn diagram for the symmetric difference of the sets A and B.	2	3	2	1
1. d)	Show that the matrix A is equal $(A^T)^T$.	2	2	2	1
1. e)	Define inductive hypothesis?	2	1	3	1
1. f)	How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?	2	4	3	1,2
1. g)	State the Bayes' theorem.	2	2	4	1
1. h)	What is the generating function for the sequence 1, 1, 1, 1, 1, 1?	2	3	4	1,3
1. i)	Name and Draw the following Graphs. K_5, C_5, W_5, $K_{3,3}$.	2	3	5	1,3
1. j)	What are the Tree Traversal Algorithms?	2	2	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Construct a truth table for each of these compound proposition $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$.	5	1	1	1,3
2. b)	Show that $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ are not logically equivalent.	5	2	1	1,2
	OR				
2. c)	Show that $\neg\forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.	5	1	1	1,2
2. d)	Show that the premises $(P \wedge Q) \vee R$ and $R \rightarrow S$ imply the conclusion $(P \vee S)$.	5	1	1	1,2
3. a)	Find the Boolean product of A and B, where $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.	5	1	2	1
3. b)	Define and produce the Cartesian product A × B × C, where A = {0, 1}, B = {1, 2}, and C = {0, 1, 2}?	5	2	2	1
	OR				
3. c)	Show that the "divides" relation is the set of positive integers in not an equivalence relation.	5	1	2	1,2
3. d)	Draw the Hasse diagram representing the partial ordering {(a, b)|a divides b} on {1, 2, 3, 4, 6, 8, 12}.	5	3	2	1,3

P.T.O.

4. a)	Show that if n is a positive integer, then $1+2+\cdots+n=\frac{n(n+1)}{2}$.	5	1	3	1,3
4. b)	Express the linear search algorithm as a recursive procedure.	5	2	3	1,2
	OR				
4. c)	What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x-3y)^{25}$?	5	3	3	1,2
4. d)	How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have, where x_1, x_2 and x_3 are non-negative integers?	5	4	3	1,2
5. a)	A sequence of 10 bits is randomly generated. What is the probability that at least one of these bits is 0?	5	4	4	1,2
5. b)	What is the variance of the random variable X, where X is the number that comes up when a fair die is rolled?	5	2	4	1,2
	OR				
5. c)	Solve the recurrence relation $a_n - 7a_{n-1} + 10\,a_{n-2} = 0\ n \geq 2$.	5	3	4	1,2
5. d)	What is the solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$?	5	3	4	1,2
6. a)	State and Prove Euler's Formula.	5	2	5	1
6. b)	Draw the minimum spanning tree for the following graph by using kruskal's algorithm. 	5	3	5	1,2,3
	OR				
6. c)	Find a minimum spanning tree for the given weighted graph using Prim's algorithm. 	5	3	5	1,2,3
6. d)	Find the minimum number of colors that are required to assign vertices of the following graph. 	5	2	5	1,2,3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome